

0.1 Ideal

Definition 0.1.1. Let R be a Ring. A subset $I \subseteq R$ is called *ideal* of R if:

1. $I \subseteq R$ is a subgroup of R .
2. I is closed under the multiplication.
3. For any $r \in R$, $rI \subseteq I$ and $Ir \subseteq I$. (In other word, for any $r \in R, a \in I$, $ra \in I$ and $ar \in I$.)

Theorem 0.1.2. Let R be a Ring. Then, TFAE:

1. $I \subseteq R$ is an Ideal of R .
2. The additive Quotient Group $R/I \stackrel{\text{def}}{=} \{r + I \mid r \in R\}$ be a Ring under the operation:

$$(r + I) \times (s + I) = (rs) + I$$

Proof. Observation:

$$r_1 + I = r_2 + I \iff r_1 - r_2 \in I \iff \exists a \in I \text{ s.t. } r_1 = r_2 + a$$

Now, for well-definedness, want to show that the equality

$$\begin{aligned} (r + I) \times (s + I) &= (rs) + I \\ &\stackrel{(*)}{=} [(r + \alpha) + I] \times [(s + \beta) + I] = (r + \alpha)(s + \beta) + I = (rs + r\beta + \alpha s + \alpha\beta) + I \end{aligned}$$

(*) holds for any $r, s \in R$, $\alpha, \beta \in I$.

If I is Ideal, then $r\beta, \alpha s, \alpha\beta \in I$. Thus closed under the addition gives (*).

Conversely, if this operation is well-defined, then for any $r, s \in R$, $\alpha, \beta \in I$, (*) holds.

Substituting zero to each r, s, α, β gives I is ideal. □

0.1.1 Properties of Ideal in Ring with identity

Definition 0.1.3. Let R be a Ring with identity, and $A \subseteq R$. Define *Ideal generated by A* as:

$$(A) \stackrel{\text{def}}{=} \bigcap_{\substack{I \text{ ideal} \\ A \subseteq I}} I$$

And,

$$\begin{aligned} RA &\stackrel{\text{def}}{=} \{r_1 a_1 + \cdots + r_n a_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\} \\ AR &\stackrel{\text{def}}{=} \{a_1 r_1 + \cdots + a_n r_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\} \\ RAR &\stackrel{\text{def}}{=} \{r_1 a_1 r'_1 + \cdots + r_n a_n r'_n \mid n \in \mathbb{N}, r_i, r'_i \in R, a_i \in A\} \end{aligned}$$

Lemma 0.1.4. Let R be a Ring with identity, and $A \subseteq R$. Then, $(A) = RAR$.

Proof. Since RAR is ideal which contains A , $(A) \subseteq RAR$.

And, conversely, if $\sum_{i=1}^n r_i a_i r'_i \in RAR$, then $\sum_{i=1}^n r_i a_i r'_i \in (A)$ because each $r_i a_i r'_i$ are contained in (A) , being (A) is ideal containing A and ideal is closed under the addition. □

Theorem 0.1.5. Let I be an ideal of Ring R with identity.

$I = R$ if and only if I contains a unit.

Proof. Right direction is clear by $1 \in R = I$.

Denote $u \in I$ be a unit with $vu = 1$, and Let $r \in R$ be given. Then,

$$r = r1 = rvu \in I$$

□

Definition 0.1.6. An Ideal M of R is *Maximal ideal* if: There is no Ideal I such that $M \subsetneq I \subsetneq R$.

Theorem 0.1.7. Let R be a Ring with identity.

Then, every proper ideal $I \subsetneq R$ is contained in a maximal ideal.

Proof. □

Prop. Let $I, J \subset R$ be ideals of a Ring R with unity.

1) $I+J$ is the smallest ideal of R containing both I and J .

2) IJ is an ideal contained in $I \cap J$.

3) If R is commutative and $I+J=R$, then $IJ=I \cap J$.

Proof. The first and second, ask a toddler on the street.

Let $a \in I \cap J$. Since $1 \in R = I+J$, $1 = b+c$ for some $b \in I$, $c \in J$, so

$$a = a \cdot 1 = a \cdot (b+c) = ab + ac = \underbrace{ba}_{\in IJ} + \underbrace{ac}_{\in IJ} \in IJ.$$

Thus $I \cap J \subseteq IJ$.

35. Let I, J, K be ideals of R . (with 1)

(a) Prove that $I(J+K) = IJ + IK$ and $(I+J)K = IK + JK$.

(b) Prove that if $J \subseteq I$ then $I \cap (J+K) = J + (I \cap K)$.

Lemma 0.1.8. Let R be a commutative Ring with identity, M, P are proper ideals of R .

1. M is Maximal Ideal if and only if R/M is a field.
2. P is Prime Ideal if and only if R/M is an integral domain.

Summary: M maximal $\iff R/M$ field $\implies R/M$ integral domain $\iff M$ prime.

Proof.

M is maximal $\iff$ There is no ideal I such that $M \subsetneq I \subsetneq R$
 $\iff$ There are Ideals of R/M only 0 and R/M
 $\iff R/M$ is field

P is Prime Ideal $\iff$ If $ab \in P$, then $a \in P$ or $b \in P$
 $\iff$ If $ab + P = P$, then $a + P = P$ or $b + P = P$
 $\iff$ If $\bar{a}\bar{b} = \bar{0}$, then $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$
 $\iff R/P$ is integral domain

□